

CHAPTER 6

Knowledge Graphs as the Memory of Validation

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Why Knowledge Graph Instead of Just Text?

The original text remains the evidence. The graph makes that evidence easier to find and connect.

TEXT

- Facts are buried in paragraphs, tables and result files.
- Relationships between facts are usually implicit.
- Finding the right evidence depends on locating the right passage.

KNOWLEDGE GRAPH

Important entities and facts are represented explicitly.

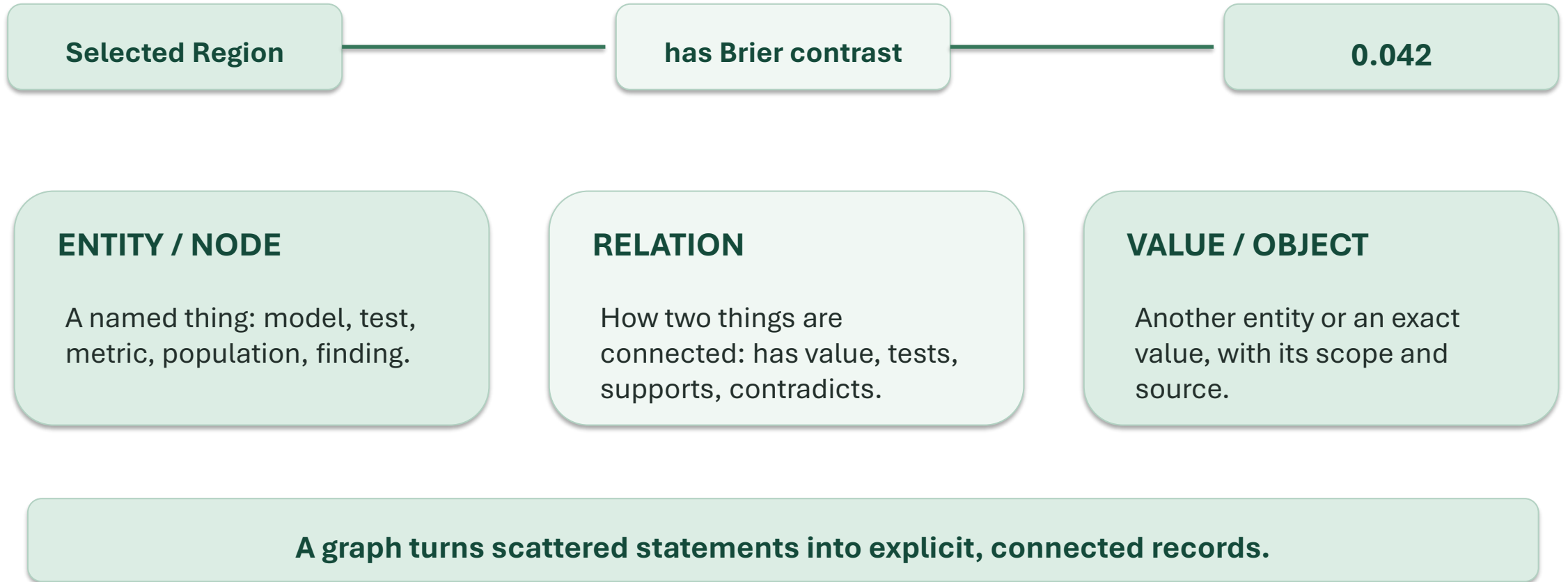
Relationships between them are recorded explicitly.

Facts can be queried directly, with a pointer back to the original source.

The KG does not replace the evidence. It is a structured index to it.

What Is a Knowledge Graph?

A simple way to store things and the relationships between them.



Ontology: The Vocabulary of the Graph

Before building the graph, define what kinds of things and relationships are allowed.

ENTITY TYPES

- Model, feature, metric, test, population, result, finding.
- The type tells us what an object is.

RELATION TYPES

has value, evaluated on, tests, supports, contradicts, derived from.

The relation tells us what the connection means.

The ontology gives different sources a common language.

What Goes Into the Validation Knowledge Graph?

Chapter 6 organizes the evidence already produced or collected during validation.

DOCUMENT EVIDENCE

- Requirements and standards
- Development documentation
- Methodology and assumptions

VALIDATION RESULTS

- Tests and measured results
- Discovered weaknesses
- Confirmation results and limitations

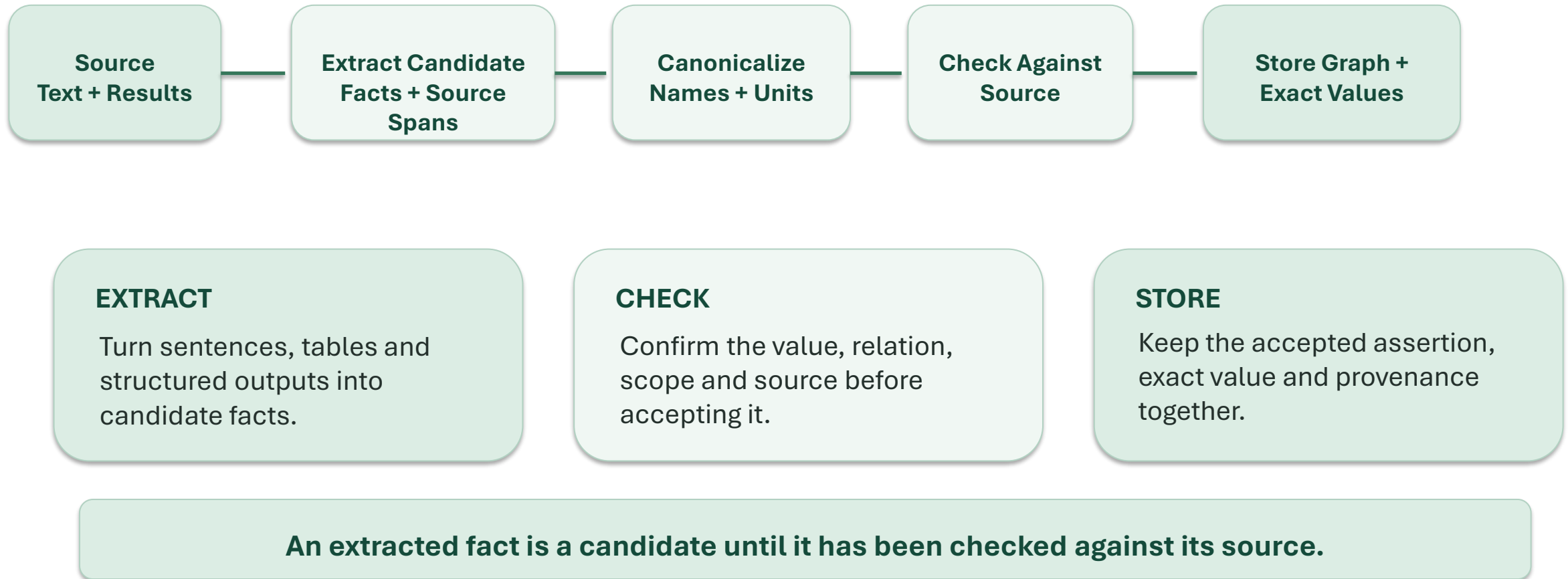
TECHNICAL RECORDS

- Model / code versions
- Data and evaluation splits
- Result files and configuration

The graph brings these records together without losing where each one came from.

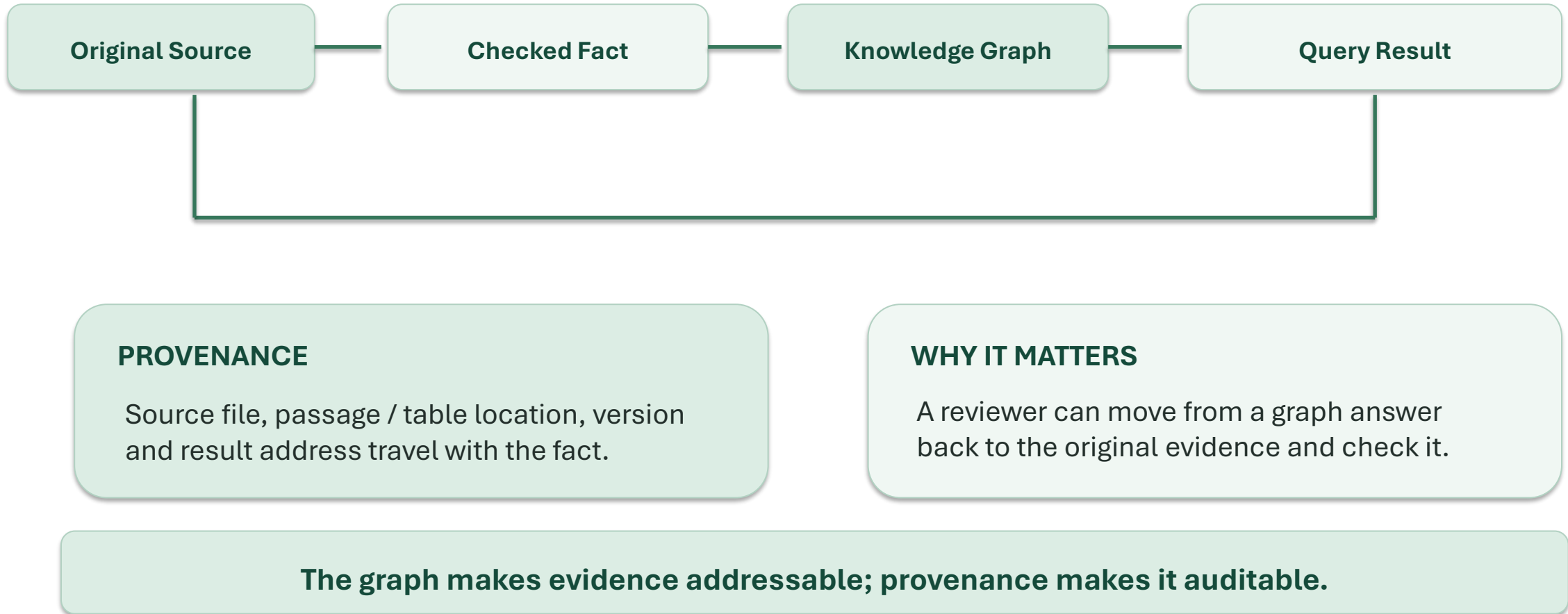
How the Knowledge Graph Is Built

Extraction proposes records; checking determines what enters the reference store.



The Graph Is an Index — The Source Remains the Evidence

Every stored fact keeps a path back to the document or result that supports it.



Three Memories, Three Different Jobs

Do not ask one representation to do everything.

SYMBOLIC MEMORY

- Entities, assertions and typed relationships.
- Best for direct graph queries and tracing connections.

NUMERICAL MEMORY

- Exact values with metric, model, population, split and configuration.
- Best for reporting numbers and deterministic comparisons.

GEOMETRIC MEMORY

- Learned representations of entities and relations.
- Useful for retrieval, compatibility and finding relationships worth examining.

Geometry can help find or rank evidence. It does not replace exact facts or source records.

How Is the Knowledge Graph Queried?

The question determines which stored records and operations are needed.

What was the confirmed loss?

Retrieve the exact numeric record + source.

What supports this finding?

Follow the finding → result → source links.

Which results concern this population?

Traverse the typed relationships.

What relationship might be worth examining?

Use geometry to rank candidates, then inspect the source-backed records.

Query the graph for the record you need, then follow provenance when the source must be inspected.